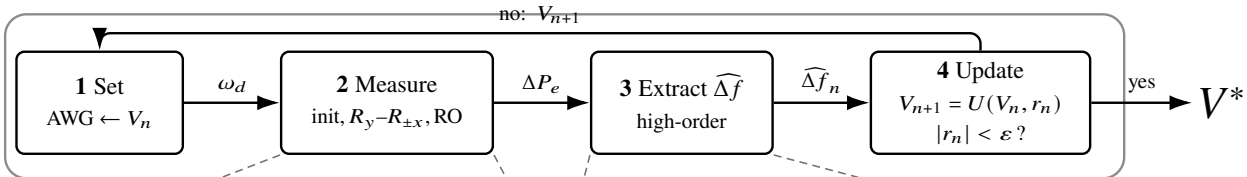
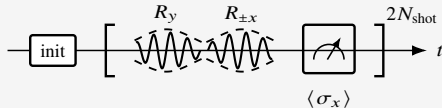


Online closed loop



Step 2: measurement sequence



order ≥ 3 extends the linear safe-zone to $|\delta \omega| \lesssim \Delta_{\text{fold}}$

Step 3: high-order extraction

$$\Delta P_e = \sum_m \int \cdots \int k_{2m+1} \times \prod_a \delta \omega(t_a) d^{2m+1}t$$

$$\downarrow \delta \omega = \text{const}$$

$$\Delta P_e = G_1 \delta \omega + \frac{1}{6} G_3 \delta \omega^3 + \cdots$$

$$\downarrow \text{Newton}$$

$$\widehat{\delta \omega} \Rightarrow \widehat{\Delta f} = \widehat{\delta \omega} / (2\pi)$$

pre-cal.

Offline calibration (once)

$$k_{2m+1} \xrightarrow{\int \cdots \int} G_{2m+1}$$